

SLASH MARK IT SOLUTIONS

Business Analyst Internship Program

Business Analyst Internship Project Report

Life Expectancy Analysis: Identifying Global Health, Healthcare and Socio-Economic Determinants Using Exploratory Data Analysis and Statistical (LASSO) Regression Modeling

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CERTIFICATE

This is to certify that Ms. Kavya P, a Bachelor of Engineering student in Computer Science and Business Systems (CSBS) from St Joseph Engineering College, Mangaluru, has successfully completed the Business Analyst Internship project titled “Life Expectancy Analysis: Identifying Global Health, Healthcare and Socio-Economic Determinants Using Exploratory Data Analysis and Statistical (LASSO) Regression Modeling” at Slash Mark IT Solutions.

The project focuses on analyzing global life expectancy data sourced from the World Health Organisation (WHO) Global Health Observatory to identify the health, healthcare-access and socio-economic factors most strongly associated with life expectancy across countries. The work involved data collection, data cleaning, missing-value treatment, outlier detection, exploratory data analysis, correlation and feature analysis, and the development of a LASSO regression model to identify the most influential predictors. The findings provide valuable analytical insights that can support evidence-based public-health and policy decision-making.

This report is submitted in partial fulfilment of the requirements for the successful completion of the Business Analyst Internship conducted by Slash Mark IT Solutions.

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I would like to express my sincere appreciation to Slash Mark IT Solutions for providing the resources, mentorship and platform that enabled me to undertake this internship project. Working on this project gave me practical exposure to real-world data cleaning, exploratory data analysis, statistical modeling and the application of data-driven thinking to a global public-health problem.

I am also grateful to my family and friends for their continuous encouragement and support throughout the course of this internship.

Completing this project has strengthened my analytical thinking, my ability to work with messy, real-world data, and my understanding of how Business Analytics techniques can be applied to support evidence-based decision-making in the public-health domain.

EXECUTIVE SUMMARY

Life expectancy is one of the most widely used summary indicators of a population's overall health and wellbeing, and is tracked by the World Health Organisation (WHO) as part of Sustainable Development Goal 3, “Good Health and Wellbeing”. Because no single measurement fully captures population health, life expectancy at birth is commonly used alongside a broad set of health, healthcare-access and socio-economic indicators to understand what drives differences in health outcomes between countries.

This project analyzed global life expectancy data using two complementary data sources built from WHO's Global Health Observatory repository: (1) a country-year panel dataset covering 193 countries and the years 2000–2015, used for in-depth exploratory data analysis, and (2) a cross-sectional dataset of national indicators for the year 2015, used to build a predictive LASSO regression model. The analysis combined data preprocessing, missing-value treatment, outlier detection (via winsorization), univariate and bivariate exploratory analysis, correlation and multicollinearity assessment, an exploratory Principal Component Analysis, and standardized LASSO regression modeling.

The exploratory analysis found that life expectancy is most strongly associated with Adult Mortality and HIV/AIDS prevalence (both negatively), and with Income Composition of Resources and Schooling (both positively). A country's development status (‘Developed’ vs ‘Developing’) was found to be a statistically significant differentiator of life expectancy (mean of 79.20 years for Developed countries versus 67.15 years for Developing countries), while the calendar year of observation was not a statistically significant driver of life expectancy differences. Following multicollinearity screening, the dataset of 20 candidate predictors was reduced to a final feature set of 12 variables for further modeling.

A separate cross-sectional dataset comprising 153 countries with complete 2015 data on twelve health, healthcare-access, environmental and socio-economic predictors was standardized and modeled using LASSO regression, evaluated across ten levels of the regularization parameter (alpha) to identify the most parsimonious model with the lowest residual sum of squares.

The findings of this project provide practical, data-driven insight into the factors most strongly associated with national life expectancy, and can support public-health organizations, governments and policy analysts in prioritizing interventions — particularly around adult mortality reduction, HIV/AIDS control, healthcare-access improvements and investment in education — in order to improve population health outcomes.

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION.....	7
1.1 Introduction.....	7
1.2 Background.....	7
CHAPTER 2: PROBLEM STATEMENT, OBJECTIVES AND SCOPE.....	9
2.1 Problem Statement.....	9
2.2 Objectives.....	9
2.3 Scope of the Project.....	10
2.4 Business Significance.....	10
CHAPTER 3: DATASET DESCRIPTION AND PROJECT METHODOLOGY.....	12
3.1 Dataset Description.....	12
3.2 Feature Description.....	13
3.3 Project Methodology.....	15
3.4 Software and Tools Used.....	16
CHAPTER 4: DATA COLLECTION AND PREPROCESSING.....	17
4.1 Introduction.....	17
4.2 Dataset Inspection.....	17
4.3 Missing Value Analysis.....	18
4.4 Outlier Detection.....	20
4.5 Feature Transformation and Data Preparation.....	22
4.6 Summary of Data Preprocessing.....	22
CHAPTER 5: EXPLORATORY DATA ANALYSIS (EDA).....	23
5.1 Introduction.....	23
5.2 Univariate Analysis of Continuous Features.....	23
5.3 Univariate Analysis of Categorical Features.....	25
5.4 Bivariate Analysis: Correlation Between Continuous Variables.....	27
5.5 Bivariate Analysis: Life Expectancy vs Year and Country Status.....	29
CHAPTER 6: STATISTICAL ANALYSIS AND PREDICTIVE MODELING.....	31
6.1 Introduction.....	31
6.2 Feature Engineering.....	31
6.3 Multicollinearity Assessment and Feature Selection.....	32
6.4 Principal Component Analysis (Exploratory).....	34
6.5 Final Feature Set.....	35
6.6 LASSO Regression Modeling.....	35
6.7 Discussion.....	38

CHAPTER 7: BUSINESS INSIGHTS AND RECOMMENDATIONS.....	39
7.1 Introduction.....	39
7.2 Key Business Insights.....	39
7.3 Business Recommendations.....	41
7.4 Business Value of the Project.....	42
CHAPTER 8: CONCLUSION, LIMITATIONS, FUTURE SCOPE AND REFERENCES.....	43
8.1 Conclusion.....	43
8.2 Limitations.....	43
8.3 Future Scope.....	44
8.4 References.....	44

CHAPTER 1: INTRODUCTION

1.1 Introduction

Life expectancy at birth is one of the most widely recognized summary indicators of a population's overall health. Because no single measurement can fully capture the health status of a country, organizations such as the World Health Organisation (WHO) rely on a broad set of health, healthcare-access, environmental and socio-economic indicators alongside life expectancy to understand what is driving differences in population health across the world.

This project analyzes global life expectancy data compiled from the WHO Global Health Observatory (GHO) repository. The analysis combines Business Analytics techniques — data cleaning, exploratory data analysis, correlation analysis and statistical modeling — to identify which health, healthcare-access and socio-economic factors are most strongly associated with life expectancy across countries, and to build a regression-based model capable of explaining variation in life expectancy using standardized national indicators.

The insights generated through this analysis can help public-health organizations, policymakers and researchers understand which categories of intervention (disease control, healthcare access, sanitation, education, or economic development) are most strongly linked to improvements in national life expectancy, and can support the prioritization of resources toward the factors with the greatest apparent association with population health outcomes.

1.2 Background

Life expectancy is monitored by the WHO as part of its efforts to implement and track the third of the United Nations' 17 Sustainable Development Goals, “Good Health and Wellbeing”. Since there is no single, universal measurement of health across the 194 WHO Member States, summary measures such as life expectancy at birth are used to provide a general assessment of a country's health situation.

A wide body of public-health research links life expectancy to factors spanning several categories: infectious disease burden (such as HIV/AIDS, measles and tuberculosis), chronic disease and lifestyle risk factors (such as alcohol consumption, being overweight, high cholesterol and high blood sugar), healthcare-system factors (such as immunization coverage and health expenditure), environmental and safety factors (such as access to improved water and sanitation, UV radiation exposure, homicide rates and road traffic deaths), and broader socio-economic factors (such as income, education and a country's overall level of development).

Understanding which of these factors are most strongly and independently associated with life expectancy, after accounting for the others, is valuable for organizations seeking to prioritize interventions with the greatest potential impact on population health. This project uses exploratory data analysis together with regularized (LASSO) regression — a technique well suited to datasets with many correlated candidate predictors — to address this question using real WHO data.

CHAPTER 2: PROBLEM STATEMENT, OBJECTIVES AND SCOPE

2.1 Problem Statement

National life expectancy varies considerably across countries, and a large number of candidate factors — spanning disease burden, healthcare access, environment and socio-economic conditions — have been proposed as drivers of this variation. Many of these candidate factors are themselves correlated with one another (for example, healthcare expenditure, immunization coverage and income are all interrelated), which makes it difficult to identify, using simple observation alone, which factors are most strongly and independently associated with life expectancy.

In addition, the underlying WHO data used for this kind of analysis is not immediately analysis-ready: it is scattered across many separate indicator tables, contains inconsistent formatting (embedded confidence intervals, non-numeric placeholder text), differing units and reporting years, and a substantial number of missing and implausible values. Before any meaningful statistical analysis can be performed, this data must be systematically collected, cleaned, merged and prepared.

This project addresses these two problems in sequence: first, by building a clean and reliable analytical dataset from WHO's public data, and second, by applying exploratory data analysis and a regularized regression technique (LASSO) to identify which health, healthcare-access and socio-economic indicators are most strongly and independently associated with national life expectancy.

2.2 Objectives

- To collect and consolidate life expectancy and a broad set of candidate predictor variables from the WHO Global Health Observatory data repository.
- To clean and preprocess the resulting dataset, including the treatment of missing values, implausible values and outliers.
- To perform exploratory data analysis (EDA) to understand the distribution of each variable and the relationships between candidate predictors and life expectancy.
- To assess correlation and multicollinearity among candidate predictors and reduce the feature set to a manageable, non-redundant set of variables.
- To evaluate dimensionality reduction via Principal Component Analysis (PCA) as an alternative feature-reduction approach.
- To build a standardized LASSO regression model to identify the predictors most strongly and independently associated with life expectancy.
- To translate the statistical findings into practical, business-relevant insights and recommendations.

2.3 Scope of the Project

The scope of this project is limited to the analysis of publicly available WHO Global Health Observatory data. Two datasets were used: a country-year panel dataset (193 countries, 2000–2015) sourced from a Kaggle-hosted, WHO-derived compilation, used for exploratory data analysis; and a bespoke cross-

sectional dataset for the year 2015, built directly from the WHO GHO API, used for LASSO regression modeling.

The analysis covers data cleaning, missing-value treatment, outlier detection, univariate and bivariate exploratory analysis, correlation and multicollinearity assessment, exploratory PCA, and standardized LASSO regression. The project does not incorporate data from outside the WHO GHO ecosystem, and does not attempt to establish causal relationships — all findings describe statistical association only, consistent with the observational, cross-country nature of the underlying data.

2.4 Business Significance

Improving population life expectancy is a central goal of public-health organizations, national governments and global health bodies such as the WHO. Because resources for public-health interventions are limited, understanding which factors are most strongly associated with life expectancy — and which of those factors are potentially actionable (such as immunization coverage or healthcare expenditure) versus largely structural (such as a country's overall development status) — is valuable for prioritizing where investment is likely to have the greatest impact.

The application of Business Analytics techniques to this problem enables a move from broad, intuitive assumptions about the drivers of health to a more systematic, data-driven understanding grounded in observed correlations and a regularized statistical model. The resulting insights can support evidence-based prioritization of health interventions, more informed international-development resource allocation, and a clearer picture of which indicators are worth tracking closely as leading signals of population health.

CHAPTER 3: DATASET DESCRIPTION AND PROJECT METHODOLOGY

3.1 Dataset Description

This project draws on two related datasets, both ultimately sourced from the WHO Global Health Observatory (GHO), used at different stages of the analysis:

3.1.1 Panel Dataset (used for Exploratory Data Analysis)

A country-year panel dataset covering 193 countries and the years 2000 to 2015 (2,938 country-year records across 22 original variables) was used for the exploratory data analysis in Chapters 4 and 5. Each row represents one country in one year and includes life expectancy together with a broad set of health, healthcare-access, environmental and socio-economic indicators.

Attribute	Description
Coverage	193 countries, years 2000–2015
Total Records	2,938 country-year rows
Total Original Variables	22 (including target and identifiers)
Target Variable	life_expectancy (years)
Data Type	Numerical and categorical (mixed)
Missing Values	Present — 16 of 22 columns contained nulls

3.1.2 Cross-Sectional Dataset (used for LASSO Regression Modeling)

A second, bespoke cross-sectional dataset was constructed directly from the WHO GHO API for the analysis year 2015 (with individual predictors sourced from the latest year with acceptably complete data, as documented per variable below). Sixteen candidate predictors were originally considered; four (Smoking, Physical Activity, Malaria and HIV incidence) were excluded due to high rates of missing data, leaving twelve predictors. After merging all indicators onto life expectancy by country and removing rows with any remaining missing values, the final analytical sample comprised 153 countries with complete data.

Attribute	Description
Coverage	153 countries with complete data (2015 or latest available year)
Total Predictors	12 (after exclusions for missingness)
Target Variable	LifeExpectancy → transformed to TransformedLife
Data Type	Numerical (standardized prior to modeling)
Extraction Date	24th July 2016 (per project documentation)

3.2 Feature Description

The table below describes the variables in the panel dataset used for exploratory data analysis.

Variable	Type	Description	Role
life_expectancy	Ratio	Life expectancy at birth, in years	Target variable
adult_mortality	Ratio	Probability of dying between ages 15–60 per 1,000 population	Predictor
infant_deaths	Ratio	Number of infant deaths per 1,000 population	Predictor
alcohol	Ratio	Per-capita alcohol consumption (litres of pure alcohol)	Predictor
percentage_expenditure	Ratio	Health expenditure as a % of GDP	Predictor
hepatitis_b	Ratio	% of 1-year-olds immunized against Hepatitis B	Predictor
measles	Ratio	Reported measles cases per 1,000 population	Predictor
bmi	Interval	Average population Body Mass Index	Dropped (~50% missing)
under-five_deaths	Ratio	Deaths of children under 5 per 1,000 population	Predictor
polio	Ratio	% of 1-year-olds immunized against Polio	Predictor
total_expenditure	Ratio	Government health expenditure as % of total government expenditure	Predictor
diphtheria	Ratio	DTP3 immunization rate among 1-year-olds	Predictor
hiv/aids	Ratio	Deaths per 1,000 live births due to HIV/AIDS (under 5s)	Predictor
gdp	Ratio	Gross Domestic Product per capita (USD)	Predictor
population	Ratio	Total country population	Predictor (later dropped)
thinness_10-19_years	Ratio	Prevalence of thinness among 10–19 year olds (%)	Predictor
thinness_5-9_years	Ratio	Prevalence of thinness among 5–9 year olds (%)	Predictor
income_composition_of_resources	Ratio	Human Development Index component (0–1 scale)	Predictor
schooling	Ratio	Average years of schooling	Predictor
status	Nominal	Country classification: 'Developed' or 'Developing'	Predictor (categorical)
country	Nominal	Country name	Identifier
year	Ordinal	Calendar year, 2000–2015	Identifier / time variable

The table below describes the twelve predictors used in the cross-sectional LASSO regression dataset, together with the measurement basis and year used for each.

Variable	Measure Used	Year
Alcohol	Litres per capita	2010
Overweight	Age-standardised % of population, BMI $\geq$ 25	2014
Cholesterol	Age-standardised % with high cholesterol	2008
BloodSugar	Crude % with high blood sugar	2014
ImprovedWater	% population using improved water sources	2015
ImprovedSanitation	% population using improved sanitation	2015
MaternalDeaths	Maternal mortality ratio per 100,000 live births	2015
UVRadiation	UV radiation exposure index	Single available year
Homicides	Homicide rate per 100,000 population	2012
RoadDeaths	Road traffic death rate per 100,000 population	2013
Tuberculosis	Tuberculosis incidence rate	2014
Suicide	Age-standardised suicide rate	2012

3.3 Project Methodology

The project followed a structured Business Analytics workflow spanning data collection, cleaning, exploratory analysis, feature engineering and statistical modeling, illustrated in the workflow below and described step by step.

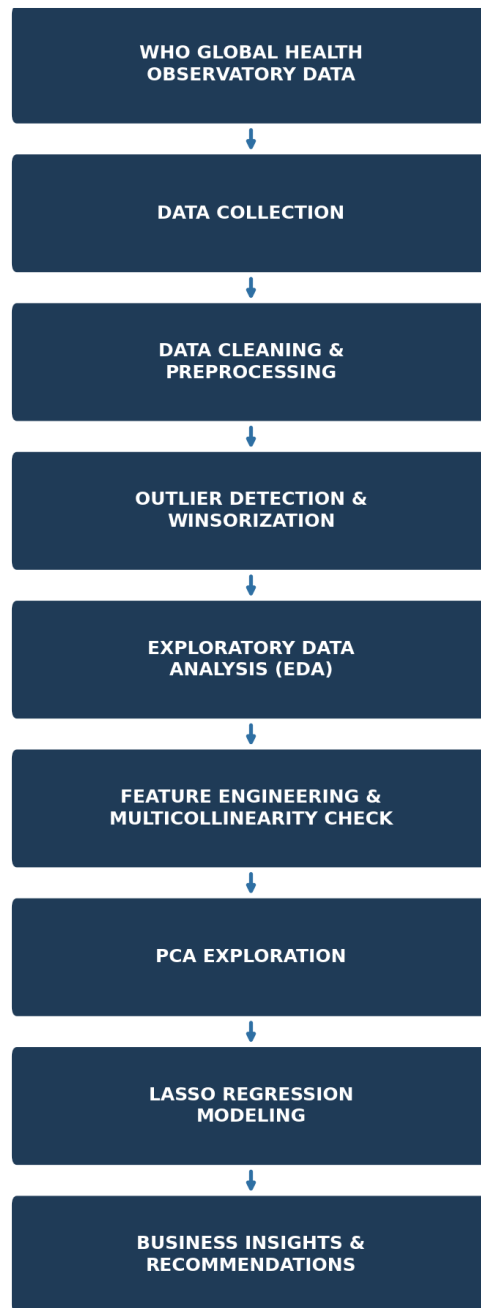


Figure 3.1: Project Methodology Workflow

Step 1: Data Collection

Life expectancy and candidate predictor data were collected from the WHO Global Health Observatory (both directly via the GHO API for the cross-sectional dataset, and via a Kaggle-hosted compilation for the panel dataset).

Step 2: Data Cleaning and Preprocessing

Column names were standardized, implausible ('inexplicit null') values were identified and converted to explicit missing values, embedded confidence-interval text was parsed out of string columns, and remaining missing values were imputed (panel dataset) or the affected rows removed (cross-sectional dataset).

Step 3: Outlier Detection and Treatment

Boxplots, histograms and Tukey's IQR method were used to detect outliers in each continuous variable of the panel dataset; outliers were treated using variable-by-variable winsorization.

Step 4: Exploratory Data Analysis

Univariate analysis (descriptive statistics, histograms, count plots) and bivariate analysis (correlation matrices, heatmaps, group comparisons and t-tests) were used to understand each variable's distribution and its relationship with life expectancy and with other predictors.

Step 5: Feature Engineering and Selection

Categorical variables were one-hot encoded, highly correlated variable pairs were identified and the less informative variable in each pair was dropped, and Principal Component Analysis was explored as an alternative dimensionality-reduction approach.

Step 6: Predictive Modeling

The cross-sectional dataset's outcome variable was transformed to correct for skew, all predictors were standardized, and a LASSO regression model was fit and evaluated across a range of regularization (alpha) values to identify the most parsimonious model.

Step 7: Business Insights and Recommendations

The statistical and exploratory findings were translated into business-relevant insights and practical recommendations for public-health prioritization.

3.4 Software and Tools Used

Tool / Library	Purpose
Python	Primary programming language for data analysis
Pandas	Data manipulation, cleaning and merging
NumPy	Numerical computation and array operations
Matplotlib	Data visualization (histograms, boxplots, line plots)
Seaborn	Statistical visualization (heatmaps, line plots)
SciPy (scipy.stats)	Statistical testing (independent-samples t-tests)
Scikit-learn	Standardization, PCA, Linear/Ridge/LASSO regression
Jupyter Notebook	Interactive analysis and literate-programming documentation
WHO GHO API	Source of life expectancy and predictor data

CHAPTER 4: DATA COLLECTION AND PREPROCESSING

4.1 Introduction

Data preprocessing is a foundational step in any data analytics project. Raw data extracted from the WHO Global Health Observatory arrives as a set of separate indicator tables, each with its own formatting quirks, embedded confidence intervals, inconsistent column names, and differing years of coverage. Before meaningful exploratory analysis or modeling could be performed, this data had to be inspected, cleaned, merged, and checked for missing values and outliers.

4.2 Dataset Inspection

The panel dataset was loaded into a Pandas DataFrame and inspected using standard techniques: viewing the first few records, examining the dimensions of the dataset (2,938 rows × 22 columns), reviewing the data type of each column, standardizing column names to a consistent lower_case_with_underscores format, and renaming the ambiguous column 'thinness_1-19_years' to the more accurate 'thinness_10-19_years'.

For the cross-sectional dataset used for LASSO modeling, each of the sixteen candidate predictor tables was inspected individually — checking data types, screening the first rows for formatting issues, and identifying columns that required numeric conversion (many indicator values were stored as text, sometimes with embedded confidence intervals such as '65.0 [63.0–67.0]').

4.3 Missing Value Analysis

4.3.1 Detecting Implausible ('Inexplicit') Missing Values

A close inspection of the panel dataset's descriptive statistics revealed a number of values that, while technically present, did not make logical sense given each variable's definition: an adult mortality rate as low as 1, infant deaths and under-five deaths reported as 0 (implausible for any real population), a Body Mass Index as low as 1 or as high as 87.3, and a GDP per capita as low as 1.68 USD. These values were treated as disguised missing data rather than genuine measurements.

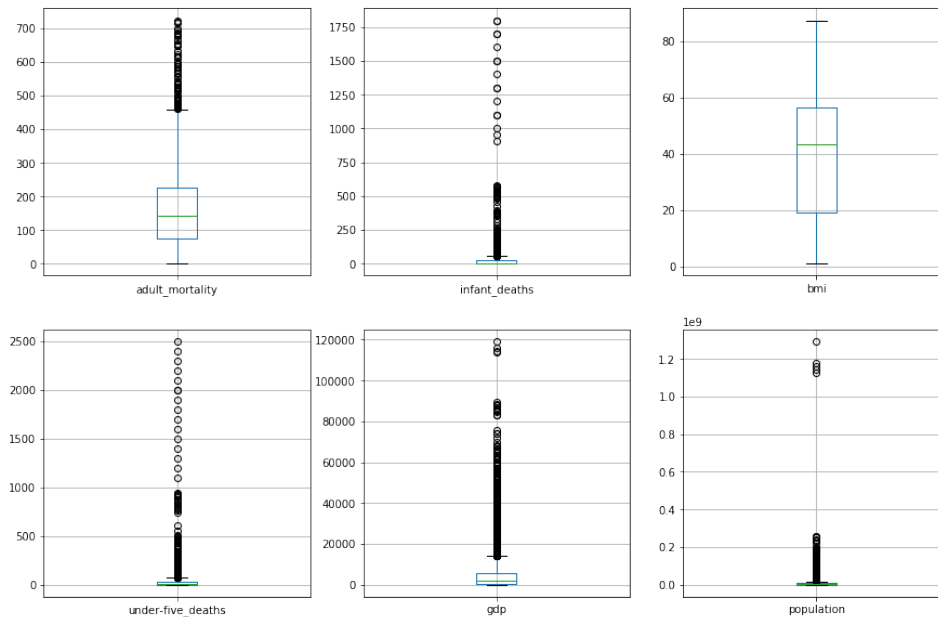


Figure 4.1: Boxplots of Candidate Error Variables (Adult Mortality, Infant Deaths, BMI, Under-Five Deaths, GDP, Population)

Observation: The boxplots for these six variables show extreme values clustered at implausible points — in particular values at or very near zero for infant deaths and under-five deaths, and BMI values well outside the physiologically realistic range of roughly 10–50.

Business Interpretation: Because these values do not represent real-world measurements, treating them as missing (rather than as valid data points or as ordinary outliers) is necessary before any statistical technique is applied; otherwise, downstream analysis would be distorted by clearly erroneous entries.

Based on this inspection, the following corrections were applied: adult mortality rates below the 5th percentile were set to missing; infant deaths values of exactly 0 were set to missing; BMI values below 10 or above 50 were set to missing; and under-five deaths values of exactly 0 were set to missing.

4.3.2 Detecting All (Explicit) Missing Values

With the implausible values converted to explicit nulls, a full null-value audit was performed across the dataset. Out of 22 total columns, 16 contained missing values (72.73% of columns). The breakdown of missing values by column is shown below.

Variable	Missing Values	% Missing
life_expectancy	10	0.34%
adult_mortality	155	5.28%
infant_deaths	848	28.86%
alcohol	194	6.60%
hepatitis_b	553	18.82%
bmi	1,456	49.56%
under-five_deaths	785	26.72%

Variable	Missing Values	% Missing
polio	19	0.65%
total_expenditure	226	7.69%
diphtheria	19	0.65%
gdp	448	15.25%
population	652	22.19%
thinness_10-19_years	34	1.16%
thinness_5-9_years	34	1.16%
income_composition_of_resources	167	5.68%
schooling	163	5.55%

4.3.3 Dealing with Missing Values

The BMI variable had by far the highest missingness, at 49.56% (1,456 of 2,938 rows), and was dropped from the dataset entirely rather than heavily imputed, since imputing half of a variable's values would risk introducing substantial noise into any downstream analysis.

Because the panel dataset is time-series data organized by country, interpolation by country was considered first; however, in practice many countries had missing values across all of their observed years for a given indicator, meaning interpolation could not fill in any values. Mean imputation by year (computing the across-country mean for each indicator in each year, and filling missing values for that year with that mean) was used instead for all remaining variables with missing data. After this step, a repeat null-value audit confirmed that 0 of the remaining 21 columns contained missing values.

4.4 Outlier Detection

With missing values resolved, each of the 18 continuous variables in the panel dataset was checked for outliers using both visual methods (boxplots and histograms) and Tukey's IQR method, where any value more than 1.5 times the interquartile range beyond the 25th or 75th percentile is flagged as an outlier.

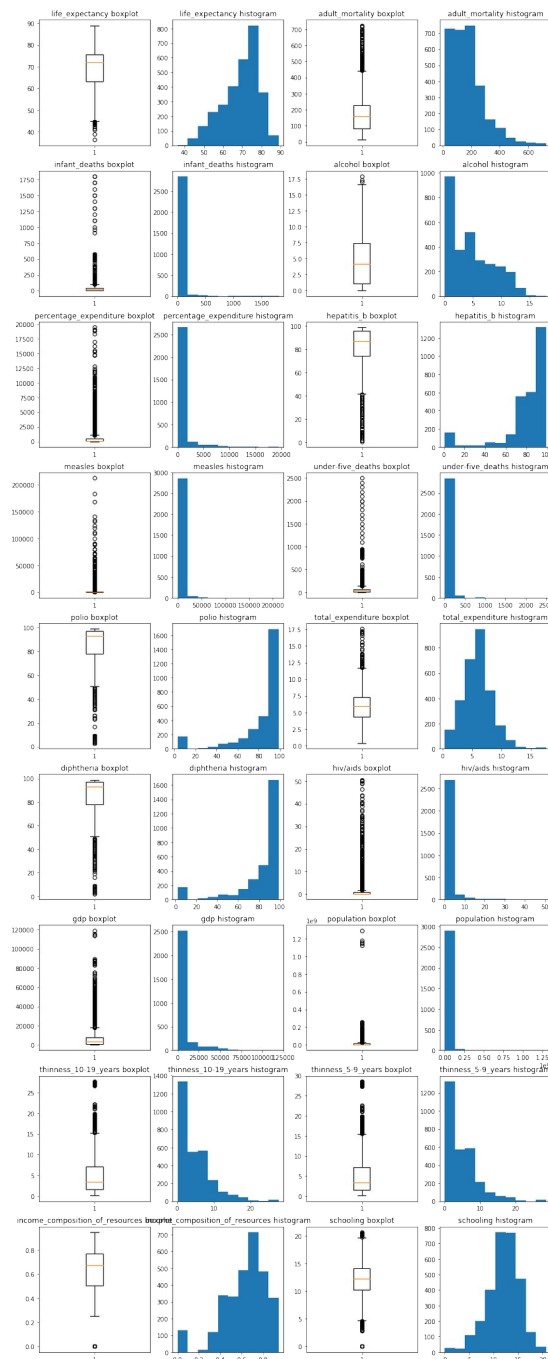


Figure 4.2: Boxplot and Histogram Grid for All Continuous Variables

Observation: Visual inspection of the boxplot/histogram pairing for each of the 18 continuous variables shows the presence of outliers in every variable, including the target variable, life expectancy. Several variables (Measles, HIV/AIDS, Percentage Expenditure) show strongly right-skewed distributions with a long tail of high-value outliers.

Business Interpretation: The presence of outliers across almost every variable is expected given the wide range of country circumstances represented in the dataset (from very small, low-income nations to very

large, high-income ones); however, left untreated, these outliers could unduly influence correlation estimates and any regression modeling performed later in the project.

The table below summarizes the number and percentage of outlier values detected in each continuous variable using Tukey's method ($1.5 \times \text{IQR}$ threshold).

Variable	Number of Outliers	% of Data
life_expectancy	17	0.58%
adult_mortality	97	3.30%
infant_deaths	135	4.59%
alcohol	3	0.10%
percentage_expenditure	389	13.24%
hepatitis_b	222	7.56%
measles	542	18.45%
under-five_deaths	142	4.83%
polio	279	9.50%
total_expenditure	51	1.74%
diphtheria	298	10.14%
hiv/aids	542	18.45%
gdp	300	10.21%
population	203	6.91%
thinness_10-19_years	100	3.40%
thinness_5-9_years	99	3.37%
income_composition_of_resources	130	4.42%
schooling	77	2.62%

Measles and HIV/AIDS show the highest outlier prevalence (542 values each, 18.45%), followed by Percentage Expenditure (389 values, 13.24%) and GDP (300 values, 10.21%). Alcohol shows the lowest outlier prevalence (3 values, 0.10%).

4.4.1 Treating Outliers: Winsorization

Because each variable had a distinct number and direction of outliers, the outliers were treated individually for each variable using winsorization — limiting extreme values to a defined percentile rather than removing the underlying rows, in order to reduce the influence of extreme values while retaining as much data as possible. For each of the 18 continuous variables, upper and/or lower winsorization limits were chosen (visually verified using before/after boxplots) so as to eliminate outliers while winsorizing as little of the data as possible.

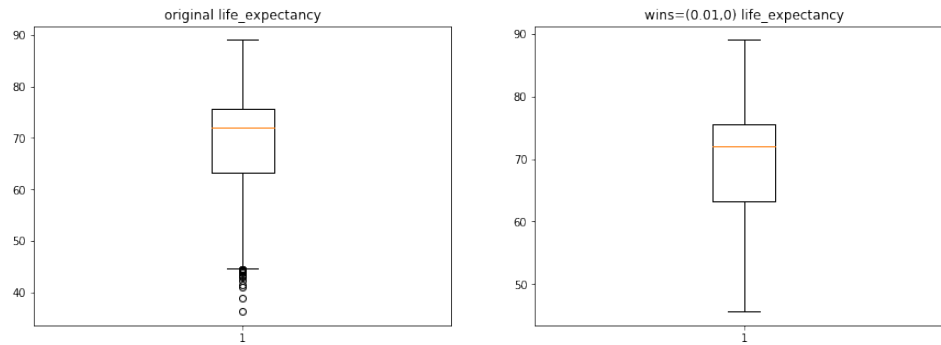


Figure 4.3: Example of Winsorization Effect (Life Expectancy, Lower Limit = 1%)

Observation: The side-by-side boxplots show the original Life Expectancy distribution alongside the winsorized version after applying a 1% lower-limit winsorization. The winsorized distribution no longer shows the extreme low-end outlier points present in the original.

Business Interpretation: Winsorizing at a modest limit (as low as 1% in this case) was sufficient to remove the extreme low outliers from Life Expectancy while preserving the overall shape and sample size of the distribution, illustrating the variable-by-variable approach used across the dataset.

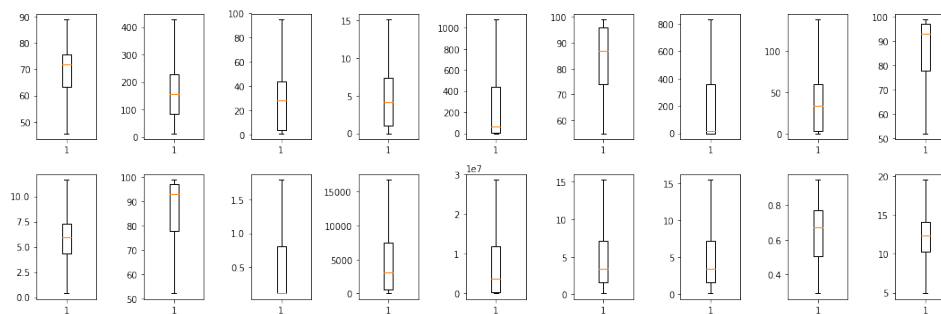


Figure 4.4: Boxplots of All Continuous Variables After Winsorization

Observation: After winsorization, the boxplots for all 18 continuous variables show a marked reduction in the number and severity of outlier points compared to the pre-treatment boxplots in Figure 4.2.

Business Interpretation: Confirming that the winsorization step was effective across all variables provided confidence that the resulting 'winsorized' dataset was suitable for use in the subsequent exploratory data analysis, without a small number of extreme values distorting summary statistics or correlation estimates.

4.5 Feature Transformation and Data Preparation

A new working DataFrame was created from the winsorized values for country, year and status alongside the treated versions of all 18 continuous variables, forming the clean dataset used throughout the exploratory analysis in Chapter 5 and the feature-engineering work in Chapter 6.

For the separate cross-sectional dataset used in LASSO regression modeling (Chapter 6), preparation followed a different but complementary path appropriate to a small, complete-case cross-sectional sample: string-formatted indicator values (some containing embedded confidence intervals in square brackets, e.g. '12.3 [10.1–14.5]') were parsed to extract the point estimate and converted to numeric type; all twelve datasets were merged onto the life expectancy table by country using an outer join; and any country with a missing value in the target or in any of the twelve predictors was removed, leaving a

complete-case sample of 153 countries. The Life Expectancy target variable was then transformed using an inverse square-root transformation to correct for negative skew (documented further in Chapter 6), and all predictor variables were standardized (zero mean, unit standard deviation) as required for LASSO regression.

4.6 Summary of Data Preprocessing

The data preprocessing phase addressed both implausible ('inexplicit') and standard missing values, treated outliers on a variable-by-variable basis using winsorization, and produced two clean, analysis-ready datasets: a 2,938-row country-year panel dataset (21 variables after dropping BMI) for exploratory analysis, and a 153-country cross-sectional dataset (12 standardized predictors) for LASSO regression modeling. These preprocessing steps established a reliable foundation for the exploratory data analysis and predictive modeling described in the following chapters.

CHAPTER 5: EXPLORATORY DATA ANALYSIS (EDA)

5.1 Introduction

With the dataset cleaned, exploratory data analysis (EDA) was performed to understand the distribution of each variable individually (univariate analysis) and the relationships between variables, including the relationship of each candidate predictor with the target variable, life expectancy (bivariate analysis). This chapter presents the key visualizations produced during this stage, along with the observations and business interpretation of each.

5.2 Univariate Analysis of Continuous Features

Descriptive statistics (mean, standard deviation, minimum, maximum and quartiles) were computed for all 18 winsorized continuous variables, followed by a full grid of histograms to visualize each variable's distribution.

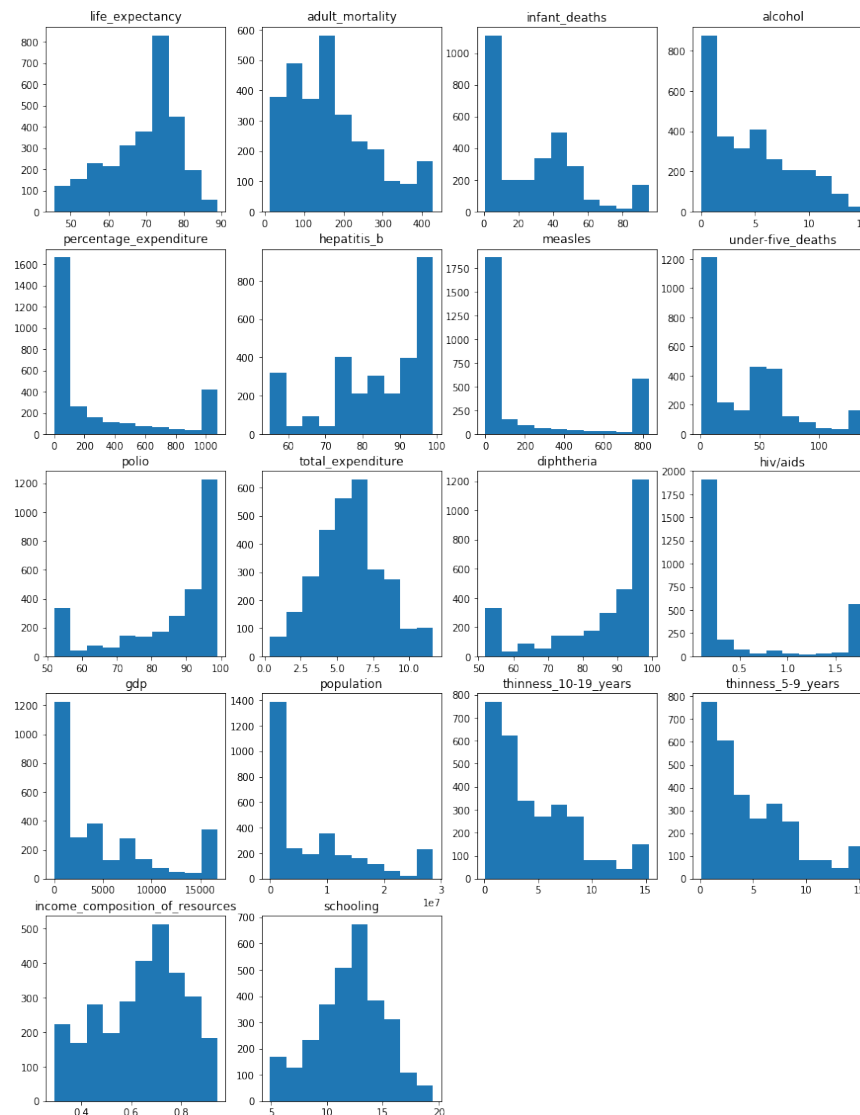


Figure 5.1: Histogram Grid of All Winsorized Continuous Variables

Observation: Winsorization had varying effects on the shape of each variable's distribution. Life Expectancy shows a left-skewed (negatively skewed) distribution concentrated in the 60–80 year range. Several health-expenditure and disease-incidence variables (GDP, Percentage Expenditure, HIV/AIDS, Measles) remain strongly right-skewed even after winsorization, reflecting genuine, very wide real-world variation between low- and high-income countries. Immunization-rate variables (Polio, Diphtheria, Hepatitis B) are heavily left-skewed, clustered near the maximum of 90–100%.

Business Interpretation: The skewed shapes of variables such as GDP and health expenditure reflect real and persistent global inequality in economic resources and healthcare investment, rather than a data-quality issue — this is useful context for interpreting the correlation and regression results later in the project, and indicates that log-style transformations could be considered for some predictors in future modeling work.

5.3 Univariate Analysis of Categorical Features

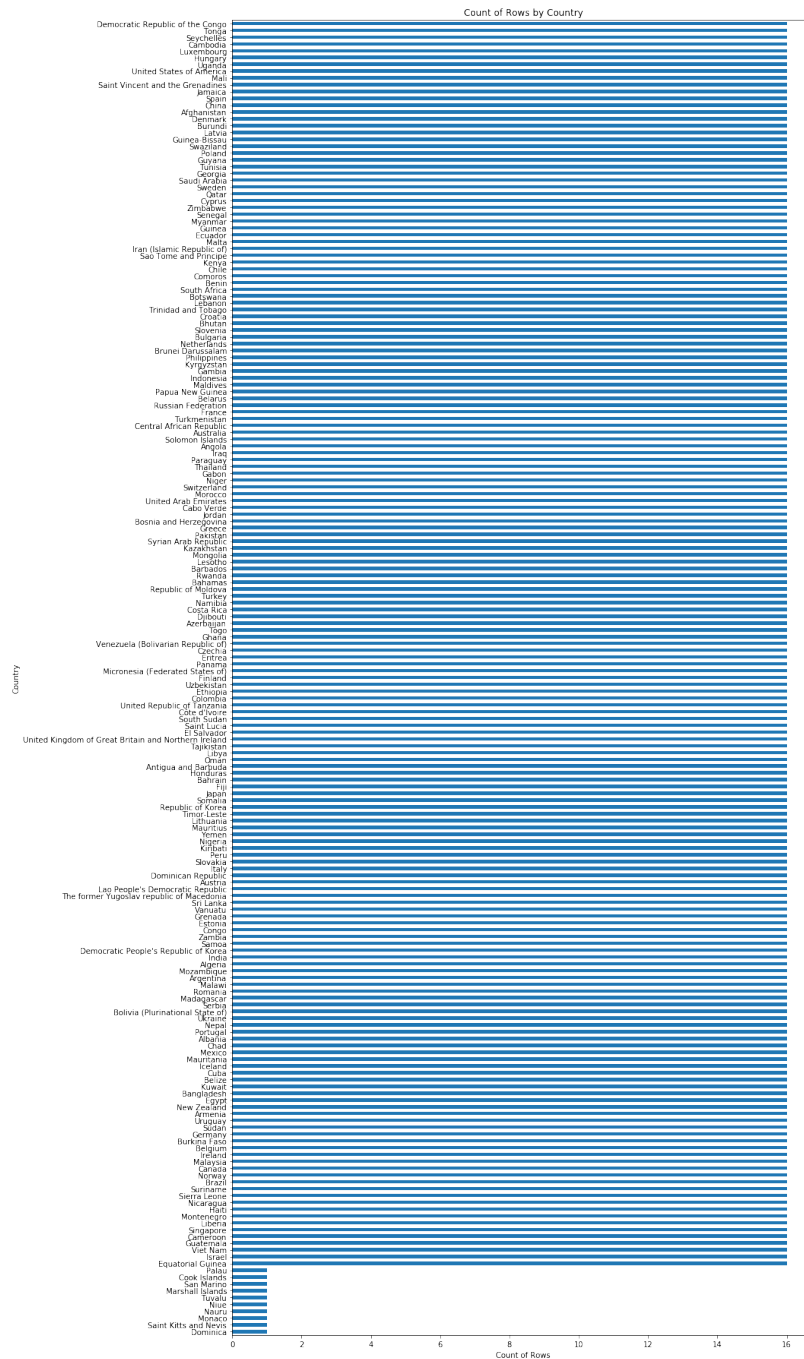


Figure 5.2: Count of Rows by Country

Observation: The vast majority of the 193 countries have 16 rows of data each (one row per year from 2000 to 2015), indicating that no individual country's data is over-represented in the panel dataset.

Business Interpretation: Confirming a roughly even distribution of observations per country provides confidence that summary statistics and correlations calculated across the full panel dataset are not being disproportionately driven by any single country having many more (or fewer) recorded years than the rest.

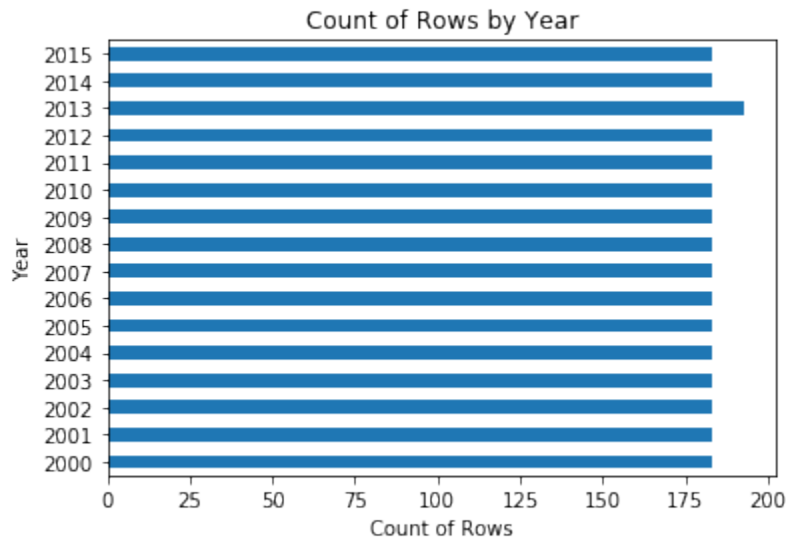


Figure 5.3: Count of Rows by Year

Observation: Each year from 2000 to 2015 has an almost identical number of rows, with the exception of 2013, which contains 10 more rows than the other years (corresponding to a small number of countries with only a single year of data, all falling in 2013).

Business Interpretation: The near-uniform distribution of rows across years means the dataset is well balanced over time and unlikely to introduce a systematic time-related bias into the exploratory or modeling results.

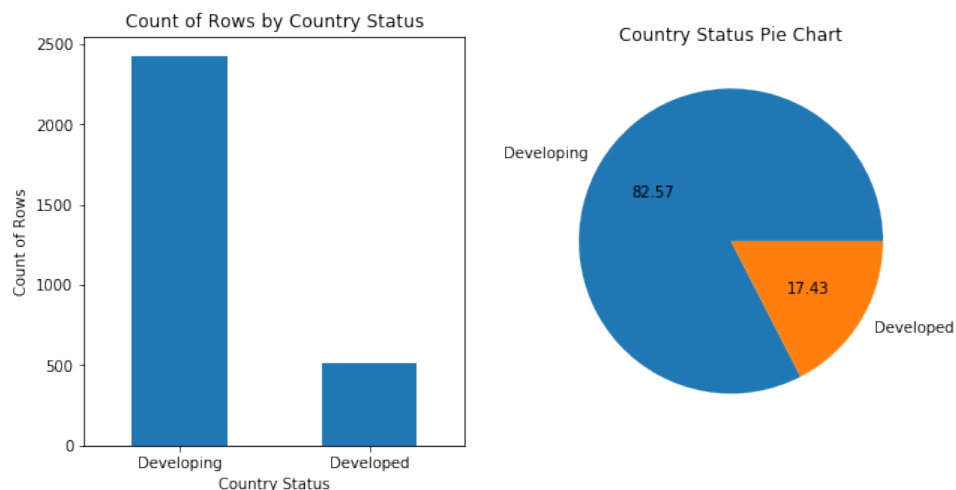


Figure 5.4: Distribution of Country Status (Bar and Pie Chart)

Observation: The bar and pie charts show that 'Developing' countries account for 2,426 of the 2,938 rows (82.57%), while 'Developed' countries account for the remaining 512 rows (17.43%).

Business Interpretation: Because the large majority of the dataset comes from 'Developing' countries, any patterns or models derived from this data are likely to more accurately reflect the dynamics of developing nations than developed ones, which is an important caveat to bear in mind when generalizing the project's findings.

5.4 Bivariate Analysis: Correlation Between Continuous Variables

A Pearson correlation matrix was computed across all 18 continuous variables, followed by a correlation heatmap to visualize the strength and direction of every pairwise relationship at a glance.

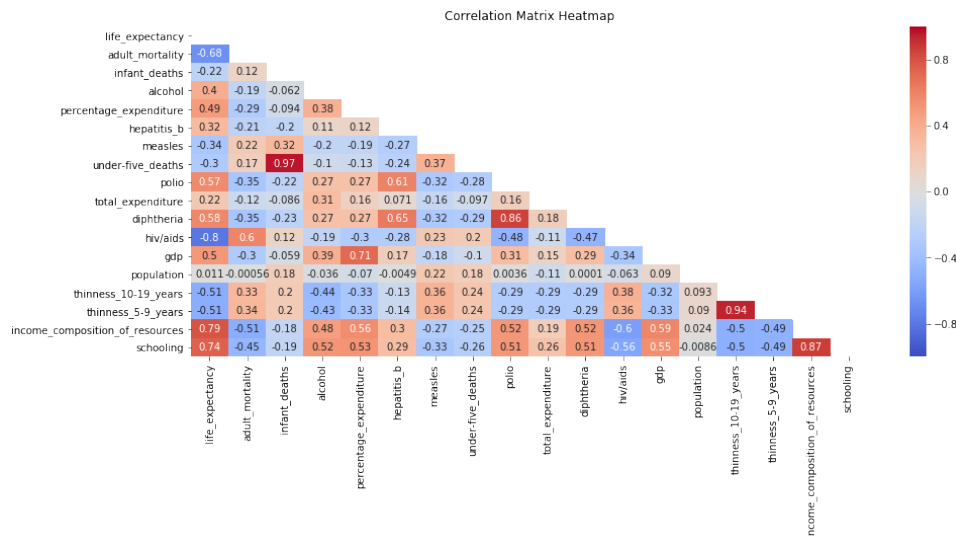


Figure 5.5: Correlation Matrix Heatmap of All Continuous Variables

Observation: The heatmap reveals several strong relationships. Life Expectancy (the target variable) is most strongly negatively correlated with HIV/AIDS ($r = -0.797$) and Adult Mortality ($r = -0.678$), and most strongly positively correlated with Income Composition of Resources ($r = 0.793$) and Schooling ($r = 0.742$). Life Expectancy is also positively correlated with Diphtheria immunization ($r = 0.579$), Polio immunization ($r = 0.573$) and GDP ($r = 0.501$), and negatively correlated with Thinness among 10–19 year olds ($r = -0.515$) and Thinness among 5–9 year olds ($r = -0.513$). Life Expectancy shows essentially no correlation with Population ($r = 0.011$). Among the predictors themselves, Infant Deaths and Under-Five Deaths are extremely highly correlated ($r = 0.967$), Percentage Expenditure and GDP are highly correlated ($r = 0.713$), Polio and Diphtheria immunization rates are highly correlated ($r = 0.856$), Thinness at 5–9 years and 10–19 years are extremely highly correlated ($r = 0.942$), and Income Composition of Resources and Schooling are highly correlated ($r = 0.874$).

Business Interpretation: The strong negative correlations with HIV/AIDS and Adult Mortality, and strong positive correlations with Income Composition of Resources and Schooling, point to disease burden and socio-economic development as the factors most closely tracking life expectancy in this dataset. At the same time, the very high correlations found between certain predictor pairs (Infant/Under-Five Deaths, Thinness at both age ranges, Income Composition/Schooling) signal a multicollinearity problem that must be addressed before building any predictive model — this motivates the feature-selection work carried out in Chapter 6.

5.5 Bivariate Analysis: Life Expectancy vs Year and Country Status

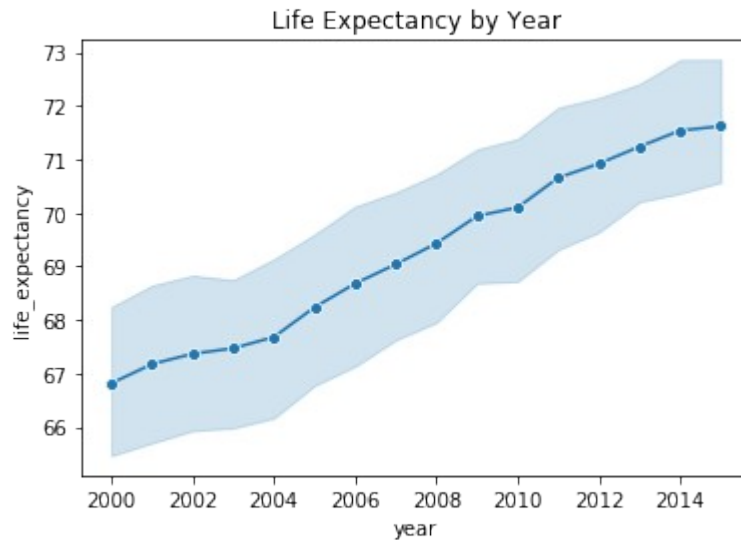


Figure 5.6: Life Expectancy Trend by Year

Observation: The line plot of average Life Expectancy by year shows a gradual upward trend from 2000 to 2015, consistent with a weak positive correlation between year and life expectancy ($r = 0.170$).

Business Interpretation: While there is a visible upward trend in average life expectancy over the fifteen-year period, the correlation is weak, and a series of year-to-year independent-samples t-tests (comparing life expectancy in each year to the following year) found no statistically significant difference between any consecutive pair of years (all p-values above 0.55). This indicates that, while the long-run trend is real, 'year' alone is not a strong enough or statistically distinct enough signal to be a useful standalone predictor and was excluded from the final feature set used for modeling in Chapter 6.

Country development status ('Developed' vs 'Developing') was also compared against Life Expectancy directly.

Status	Mean Life Expectancy (years)
Developed	79.20
Developing	67.15

An independent-samples t-test comparing these two groups ($t = 47.96$, $p \approx 2 \times 10^{-323}$) confirmed that the difference in mean life expectancy between Developed and Developing countries is highly statistically significant. Given this result, Developed and Developing countries were also compared across all other continuous variables (adult mortality, alcohol consumption, healthcare expenditure, immunization rates, HIV/AIDS, GDP, thinness, income composition of resources and schooling, among others); statistically significant differences ($p < 0.05$) were found for every one of these variables except Infant Deaths ($p = 0.437$), confirming that a country's development status is closely intertwined with almost every health and socio-economic indicator in the dataset, and should be included as a feature in the predictive model developed in Chapter 6.

Chapter Summary

The exploratory analysis identified a clear set of variables most strongly associated with life expectancy — in particular HIV/AIDS, Adult Mortality, Income Composition of Resources and Schooling — while also surfacing several groups of highly inter-correlated predictors (infant/under-five deaths, the two thinness measures, and income composition/schooling) and confirming that country development status, but not calendar year, is a statistically significant differentiator of life expectancy. These findings directly inform the feature engineering and modeling work described in Chapter 6.

CHAPTER 6: STATISTICAL ANALYSIS AND PREDICTIVE MODELING

6.1 Introduction

Building on the exploratory findings from Chapter 5, this chapter describes the feature-engineering process used to reduce the panel dataset's variables down to a compact, non-redundant set, the exploratory use of Principal Component Analysis (PCA) as an alternative dimensionality-reduction technique, and the LASSO regression model built on the separate, standardized cross-sectional dataset to identify the predictors most strongly and independently associated with life expectancy.

6.2 Feature Engineering

Since the exploratory analysis in Chapter 5 established that a country's development status is significantly associated with life expectancy, the categorical 'status' variable was one-hot encoded into two indicator columns, 'Developed' and 'Developing'. The categorical 'country' and 'year' columns were dropped from the feature set, since Chapter 5's analysis found no statistically significant year-to-year differences in life expectancy, and country has too many unique values with too few observations each to be useful as a categorical predictor in a regression context.

6.3 Multicollinearity Assessment and Feature Selection

A correlation heatmap was recomputed on the resulting feature set (including the new 'Developed'/'Developing' indicators) to identify pairs of predictors that were very highly or extremely highly correlated with one another (defined as $|r| > 0.7$).

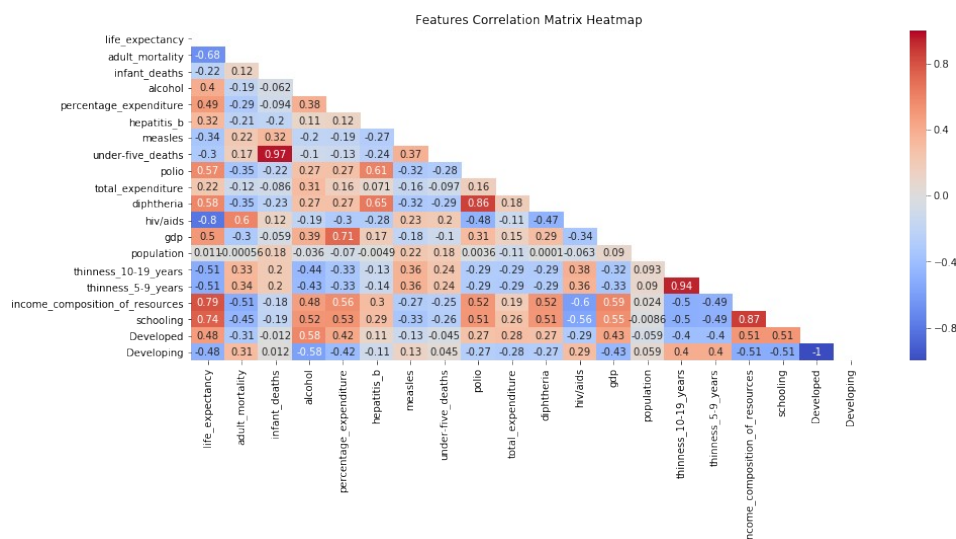


Figure 6.1: Feature Correlation Matrix Heatmap (After One-Hot Encoding of Status)

Observation: The 'Developed' indicator correlates moderately with several predictors (most notably GDP, Income Composition of Resources and Schooling) but shows very low correlation with Infant Deaths, Under-Five Deaths and Population.

Business Interpretation: The moderate-but-not-extreme correlation between 'Developed' status and most other predictors suggests that development status captures useful, complementary information rather than being redundant with any single existing predictor, supporting its inclusion in the final feature set.

The following highly correlated variable pairs ($|r| > 0.7$) were identified, and in each case the variable more weakly correlated with Life Expectancy was dropped from the feature set:

- Infant Deaths / Under-Five Deaths ($r = 0.967$) — Infant Deaths dropped; Under-Five Deaths is more highly correlated with Life Expectancy.
- GDP / Percentage Expenditure ($r = 0.713$) — Percentage Expenditure dropped; GDP is more highly correlated with Life Expectancy.
- Polio / Diphtheria ($r = 0.856$) — Polio dropped; Diphtheria is more highly correlated with Life Expectancy.
- Thinness 5-9 Years / Thinness 10-19 Years ($r = 0.942$) — Thinness 10-19 Years dropped, as its correlations with other variables were slightly higher (more redundant).
- Income Composition of Resources / Schooling ($r = 0.874$) — Schooling dropped; Income Composition of Resources is more highly correlated with Life Expectancy.
- Developed / Developing ($r = -1.0$, by construction) — Developing dropped, as it is the exact complement of Developed.

In addition to these highly correlated pairs, the Population variable was dropped from the feature set, as it showed very low correlation with almost every other variable in the dataset (including the target), and therefore contributed little explanatory information.

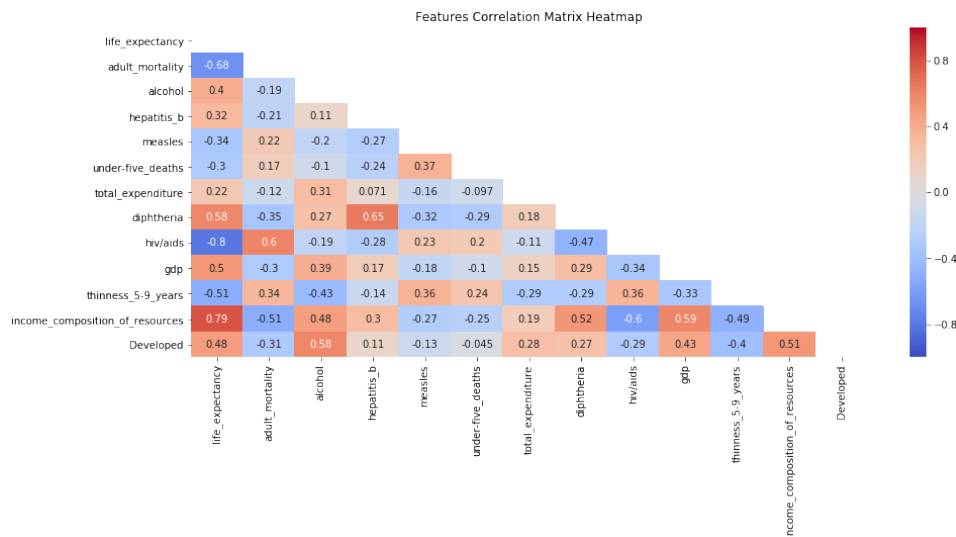


Figure 6.2: Final Feature Correlation Matrix Heatmap (After Multicollinearity Reduction)

Observation: With the redundant and weakly informative variables removed, the resulting correlation heatmap shows a more balanced structure, with no remaining pairs of predictors at the extreme correlation levels seen previously.

Business Interpretation: This reduced feature set retains the variables most informative about Life Expectancy while minimizing redundancy between predictors, which is important both for the

interpretability of a regression model's coefficients and for the reliability of any feature-reduction technique (such as the PCA explored next) applied to the data.

6.4 Principal Component Analysis (Exploratory)

Principal Component Analysis (PCA) was explored as an alternative, unsupervised approach to reducing the dimensionality of the feature set. PCA requires several assumptions to hold for best results: continuous data, an adequate sample-to-feature ratio, approximately normal and linearly related variables, meaningful correlation between features, and the absence of outliers. The reduced feature set (with the 'Developed' indicator and the target variable, Life Expectancy, removed, since PCA is unsupervised) satisfied the sample-size, correlation and outlier-related assumptions, though the linearity and normality assumptions were less certain to hold given the skewed distributions observed in Chapter 5.

PCA was fit on the eleven remaining continuous predictors (standardized prior to fitting). The results are summarized below.

Metric	Result
Number of input features	11
PC1 explained variance	36.69%
Components needed for $\geq 90\%$ variance	8 of 11
Components with eigenvalue ≥ 1	3

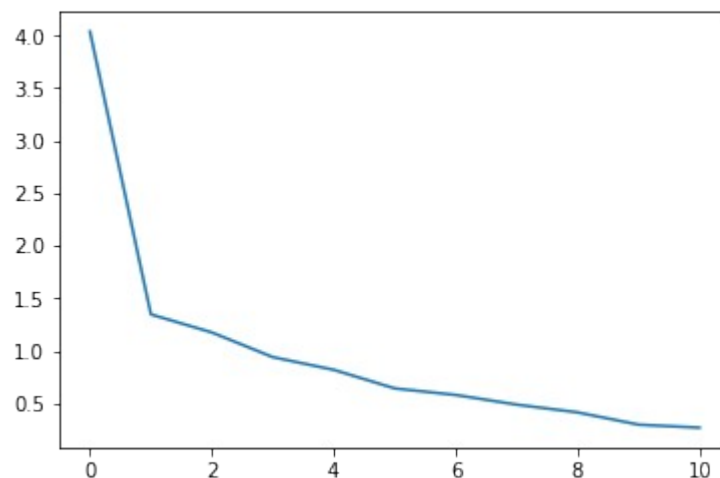


Figure 6.3: PCA Scree Plot (Explained Variance by Component)

Observation: The scree plot shows a sharp drop after the first principal component, which alone explains 36.69% of total variance (eigenvalue 4.04). Only three components have an eigenvalue at or above 1, and eight of the eleven components are required to capture at least 90% of total variance.

Business Interpretation: Because capturing 90% of the variance in this feature set would require dropping only three of the eleven original features, PCA does not offer a meaningful dimensionality-reduction benefit here — the loss of interpretability that comes from working with abstract principal components instead of named health and socio-economic indicators is not justified by a correspondingly large reduction

in the number of features. For this reason, the project proceeded with the pre-PCA feature set rather than principal components.

6.5 Final Feature Set

Following the multicollinearity-based feature selection described in Section 6.3, the panel dataset's feature set was reduced from 20 original candidate predictors (excluding country and year) down to 12 final features used to describe variation in Life Expectancy:

- Adult Mortality
- Alcohol
- Hepatitis B
- Measles
- Under-Five Deaths
- Total Expenditure
- Diphtheria
- HIV/AIDS
- GDP
- Thinness 5-9 Years
- Income Composition of Resources
- Developed (status indicator)

6.6 LASSO Regression Modeling

A separate cross-sectional dataset (153 countries, 2015 or latest available year, 12 predictors — see Section 3.1.2) was used to build a predictive regression model of Life Expectancy. LASSO (Least Absolute Shrinkage and Selection Operator) regression was chosen over ordinary linear regression and ridge regression specifically because of its ability to shrink less-informative predictor coefficients toward zero, effectively performing automatic feature selection — an important property given the correlated nature of many of the candidate predictors, as also observed in the panel-dataset analysis above.

6.6.1 Outcome Transformation

A histogram of the raw Life Expectancy variable in the cross-sectional dataset showed a pronounced negative skew (a long left tail driven by a small number of countries with much lower life expectancy than the rest). To correct this, an inverse square-root transformation was applied: $\text{TransformedLife} = \sqrt{(\max(\text{LifeExpectancy}) + 1) - \text{LifeExpectancy}}$. Because this transformation inverts the scale, a higher TransformedLife value corresponds to a lower (raw) life expectancy, and a lower TransformedLife value corresponds to a higher life expectancy — this must be kept in mind when interpreting the sign of any regression coefficient fit against TransformedLife, and predictions must be back-transformed to be interpreted on the original life-expectancy scale.

6.6.2 Predictor Standardization

As recommended practice for LASSO regression (so that the size of the regularization penalty is applied comparably across predictors measured on very different scales, such as a 0–100% immunization rate versus a per-100,000-population homicide rate), all twelve predictors were standardized to zero mean and unit standard deviation prior to model fitting.

6.6.3 Alpha (Regularization Strength) Selection

The LASSO model was evaluated across ten levels of the regularization parameter, alpha, ranging from $1e-15$ (effectively unregularized) to 10 (heavily regularized), fitting a separate model at each alpha and recording the resulting residual sum of squares (RSS) and coefficient values. This approach follows a standard methodology for selecting a regularization strength that balances model fit against model simplicity (parsimony).

Alpha Value	Characteristic
$1e-15$	Effectively unregularized
$1e-10$	Effectively unregularized
$1e-8$	Very light regularization
$1e-5$	Light regularization
$1e-4$	Light regularization
$1e-3$	Selected: most parsimonious model with lowest RSS
$1e-2$	Moderate regularization — further shrinkage
1	Strong regularization
5	Very strong regularization
10	Maximum regularization tested

Based on this evaluation, an alpha value of 0.001 was selected as the most parsimonious model that still achieved the lowest residual sum of squares among the tested levels, consistent with the project's documented aim of using LASSO specifically for its feature-reduction property rather than simply maximizing in-sample fit.

6.6.4 Multicollinearity Check

Before finalizing the model, the twelve standardized predictors were also screened for multicollinearity: two predictor pairs were found to have a correlation above 0.8 (Improved Sanitation with Improved Water, and Improved Sanitation with Cholesterol). A sensitivity analysis — comparing model coefficients with and without these variables included together — found that including both members of each pair alongside one another did not materially change the estimated coefficients, so both variables in each pair were retained in the final model, consistent with LASSO's own ability to shrink and manage correlated predictors.

6.6.5 Working Model Coefficients

The project's working scripts document a fitted model expressing predicted life expectancy as a function of the standardized predictor set (an eleven-predictor iteration of the model, prior to the final twelve-predictor / $\alpha = 0.001$ configuration described above), presented here with and without an intercept term:

Without intercept:

$$E(Y|X) = 0 + 0.053(\text{Alcohol}) - 0.36(\text{Cholesterol}) + 0.22(\text{BloodSugar}) - 0.067(\text{ImprovedWater}) - 0.22(\text{ImprovedSanitation}) + 0.27(\text{MaternalDeaths}) + 0.00056(\text{UVRadiation}) + 0.011(\text{Homicides}) + 0.26(\text{RoadDeaths}) + 0.23(\text{Tuberculosis}) + 0.047(\text{Suicide})$$

With intercept:

$$E(Y|X) = 3.4 + 0.045(\text{Alcohol}) - 0.36(\text{Cholesterol}) + 0.22(\text{BloodSugar}) - 0.065(\text{ImprovedWater}) - 0.22(\text{ImprovedSanitation}) + 0.26(\text{MaternalDeaths}) + 0.01(\text{Homicides}) + 0.26(\text{RoadDeaths}) + 0.23(\text{Tuberculosis}) + 0.045(\text{Suicide})$$

Because these coefficients are fit against the transformed outcome, TransformedLife (which moves inversely with actual life expectancy), a positive coefficient indicates a variable associated with lower life expectancy, while a negative coefficient indicates a variable associated with higher life expectancy. Under this interpretation, Alcohol consumption, Maternal Deaths, Homicides, Road Deaths, Tuberculosis incidence and Suicide rate are all associated with lower predicted life expectancy, while Cholesterol and Improved Sanitation/Water access are associated with higher predicted life expectancy in this fitted model. The counter-intuitive positive association between Cholesterol and life expectancy documented here most likely reflects the fact that high-cholesterol prevalence tends to be reported more in wealthier countries with more comprehensive health screening and generally higher life expectancy, illustrating the difference between statistical association and causation in an observational, cross-country dataset of this kind.

6.7 Discussion

Both strands of statistical analysis — the panel-dataset correlation and feature-selection work, and the cross-sectional LASSO regression — point toward a consistent overall picture: mortality-related and disease-burden indicators (Adult Mortality, HIV/AIDS, Maternal Deaths, Tuberculosis) and socio-economic indicators (Income Composition of Resources, GDP, and, in the working LASSO model, sanitation and water access) are the variables most consistently associated with life expectancy across the data explored in this project.

It should be emphasized that all reported LASSO model coefficients and the intermediate alpha-selection results reflect the values and methodology explicitly documented in the project's own analysis scripts; because the underlying WHO API calls depend on a live, external data source, the project's notebooks were not re-executed as part of this report, and no additional coefficient values, accuracy metrics, or model performance figures beyond those explicitly documented in the project code have been included. The cross-sectional model was also fit and evaluated entirely on its own training data (all 153 countries), with no held-out test set, given the modest sample size — as noted in the project's own documentation, this means the reported fit is likely optimistic relative to how the model would perform on genuinely unseen country data, and should be treated as an in-sample, explanatory result rather than a validated out-of-sample predictive one.

CHAPTER 7: BUSINESS INSIGHTS AND RECOMMENDATIONS

7.1 Introduction

The primary objective of Business Analytics is not only to analyze data but also to translate analytical findings into actionable insight. Having performed exploratory data analysis, correlation and multicollinearity assessment, and LASSO regression modeling, this chapter summarizes the key business insights that emerged from the project and the practical recommendations that follow from them.

7.2 Key Business Insights

Insight 1: Disease Burden Is the Strongest Driver of Life Expectancy Differences

Observation: HIV/AIDS ($r = -0.797$) and Adult Mortality ($r = -0.678$) were the two most strongly correlated variables with Life Expectancy in the panel dataset, both negatively.

Business Insight: Efforts to reduce adult mortality and control HIV/AIDS prevalence are likely to have an outsized association with improvements in national life expectancy relative to most other factors examined. Public-health investment aimed at disease prevention and adult-mortality reduction should be treated as a high-priority lever.

Insight 2: Socio-Economic Development Is Nearly as Strongly Associated with Life Expectancy as Disease Burden

Observation: Income Composition of Resources ($r = 0.793$) and Schooling ($r = 0.742$) were the two most strongly positively correlated variables with Life Expectancy.

Business Insight: Investment in education and broader economic development is closely intertwined with population health outcomes. Because Income Composition of Resources and Schooling were themselves highly correlated with one another ($r = 0.874$), this points to a broader, unified 'human development' factor — rather than either variable in isolation — being what tracks life expectancy most closely.

Insight 3: A Country's Development Status Is a Powerful, Statistically Significant Differentiator

Observation: Developed countries showed a mean life expectancy of 79.20 years versus 67.15 years for Developing countries, a highly statistically significant difference ($p \approx 2 \times 10^{-323}$).

Business Insight: The roughly 12-year life-expectancy gap between Developed and Developing countries is the single largest, clearest pattern identified in this project. It reflects the cumulative effect of the many underlying factors (healthcare access, disease burden, income, education) that were also individually found to differ significantly between the two groups — all but one (Infant Deaths) of the continuous variables examined showed statistically significant differences by development status.

Insight 4: Calendar Year Alone Is Not a Meaningful Predictor of Life Expectancy Differences

Observation: While average life expectancy showed a gradual upward trend from 2000 to 2015, the year-to-year correlation was weak ($r = 0.170$) and no consecutive pair of years showed a statistically significant difference in life expectancy (all p-values above 0.55).

Business Insight: Long-run global health improvements are real but occur gradually; year-over-year monitoring alone is unlikely to reveal actionable short-term signals, and analytical effort is better directed toward the underlying health and socio-economic drivers identified in this project rather than toward year-based trend-fitting.

Insight 5: Several Predictor Groups Carry Largely Redundant Information

Observation: Infant Deaths and Under-Five Deaths ($r = 0.967$), the two Thinness measures ($r = 0.942$), Polio and Diphtheria immunization ($r = 0.856$), and Income Composition of Resources and Schooling ($r = 0.874$) were all found to be extremely highly correlated with one another.

Business Insight: Organizations tracking these indicators for monitoring purposes could reasonably streamline their reporting by focusing on one representative indicator from each highly correlated group, without a meaningful loss of information — this project retained Under-Five Deaths, Thinness at 5–9 Years, Diphtheria, and Income Composition of Resources as the more informative member of each respective pair.

Insight 6: Lifestyle, Environmental and Safety Factors Show a Consistent, if More Modest, Association with Life Expectancy

Observation: In the working LASSO regression model, Alcohol consumption, Maternal Deaths, Homicide rate, Road Traffic Deaths, Tuberculosis incidence and Suicide rate were all associated with lower predicted life expectancy, while access to improved water and sanitation was associated with higher predicted life expectancy.

Business Insight: Beyond the dominant disease-burden and development factors, a broader set of preventable-death and environmental-access factors (road safety, violence prevention, maternal health, sanitation infrastructure) also carry a consistent, directionally sensible association with life expectancy and represent additional, more specifically actionable areas for targeted public-health and infrastructure investment.

7.3 Business Recommendations

- Prioritize investment in adult-mortality reduction and HIV/AIDS prevention and treatment programs, given their outsized statistical association with life expectancy relative to other factors examined.
- Treat education access and broader income/human-development investment as a complementary, high-leverage lever for improving population health outcomes over the long term, rather than viewing health and economic development policy as separate tracks.
- When designing country-comparison dashboards or reports, retain a country's development status as a standing contextual variable, since it is significantly associated with nearly every health indicator examined.
- Streamline routine health-indicator monitoring by consolidating highly redundant indicator pairs (infant/under-five mortality, the two thinness measures, polio/diphtheria immunization, income composition/schooling) into a single representative metric per group, reducing reporting overhead without a meaningful loss of insight.

- Continue to track and invest in maternal health, sanitation and water infrastructure, road safety and violence-prevention programs as secondary, more specifically actionable levers alongside the primary disease-burden and development-focused interventions.
- Treat any regression-based projections from this analysis as descriptive of statistical association rather than proof of causation, and validate any prioritization decision with additional, ideally interventional or longitudinal, evidence before committing significant resources.

7.4 Business Value of the Project

This project demonstrates how Business Analytics techniques — systematic data cleaning, exploratory analysis, correlation and multicollinearity assessment, and regularized regression modeling — can be applied to a genuine global public-health dataset to move from broad assumptions about the drivers of life expectancy toward a specific, evidence-based understanding. The outcomes of this project enable stakeholders to:

- Identify the health, healthcare-access and socio-economic factors most strongly associated with national life expectancy.
- Prioritize public-health and development interventions using a data-driven, rather than purely intuitive, basis.
- Understand which routinely tracked indicators are redundant with one another, supporting more efficient monitoring and reporting.
- Recognize the scale of the life-expectancy gap between Developed and Developing countries and the breadth of factors underlying it.
- Apply a reusable analytical workflow (data collection, cleaning, EDA, feature selection, regularized regression) to future, similarly structured public-health datasets.

CHAPTER 8: CONCLUSION, LIMITATIONS, FUTURE SCOPE AND REFERENCES

8.1 Conclusion

Life expectancy is a critical summary indicator of population health, and understanding the factors most strongly associated with it is essential for prioritizing public-health and development interventions. In this project, life expectancy data compiled from the WHO Global Health Observatory was analyzed using Business Analytics techniques and statistical modeling methods.

The project involved data collection from the WHO GHO API and a Kaggle-hosted WHO data compilation, extensive data cleaning (treatment of implausible and missing values, outlier detection and winsorization), exploratory data analysis (univariate and bivariate techniques, correlation analysis, and independent-samples t-tests), feature engineering and multicollinearity-based feature selection, an exploratory Principal Component Analysis, and the development of a standardized LASSO regression model.

The analysis found that HIV/AIDS and Adult Mortality are the factors most strongly negatively associated with life expectancy, while Income Composition of Resources and Schooling are the factors most strongly positively associated with it. A country's development status was found to be a highly statistically significant differentiator of life expectancy, while calendar year was not. The LASSO regression modeling further identified Alcohol consumption, Maternal Deaths, Homicide rate, Road Traffic Deaths, Tuberculosis incidence and Suicide rate as associated with lower predicted life expectancy, and access to improved water and sanitation as associated with higher predicted life expectancy.

The insights obtained from this analysis can assist public-health organizations, governments and policy analysts in making informed, evidence-based decisions about where to prioritize resources and attention in order to improve population health outcomes. Overall, the project demonstrates how Business Analytics and statistical modeling can support evidence-based decision-making in the global public-health domain.

8.2 Limitations

- The analysis is observational and cross-country in nature; all reported relationships describe statistical association, not causation, and should not be interpreted as proof that changing one factor will directly change life expectancy.
- The panel dataset relies on mean-imputation by year for missing values in several variables with substantial missingness (up to ~29% for Infant Deaths and ~27% for Under-Five Deaths prior to the BMI-style exclusion), which may understate the true variability of those indicators.
- The BMI variable was dropped entirely due to a missingness rate of nearly 50%, meaning body-mass-related effects on life expectancy could not be assessed in this project.
- The cross-sectional LASSO model was fit and evaluated on all 153 available countries with no held-out test set, given the limited sample size; the reported fit is therefore an in-sample result and may not generalize as well to new data.
- The project relies on the WHO Global Health Observatory and a derived Kaggle compilation as its sole data sources, and does not incorporate other potentially relevant factors such as healthcare-system quality measures, conflict/political-stability indicators, or more granular sub-national data.

- Winsorization and imputation choices, while documented and visually verified variable by variable, involve some degree of analyst judgment that could affect the precise correlation and coefficient values reported.

8.3 Future Scope

- Re-run the WHO GHO API extraction with the most current data available and validate whether the relationships identified in this project remain stable using more recent years of data.
- Evaluate alternative and more advanced regression or machine-learning techniques (e.g. Random Forest, Gradient Boosting) alongside LASSO, with proper train/test or cross-validation splits, to assess whether predictive performance can be improved without sacrificing interpretability.
- Incorporate additional data sources beyond WHO GHO, such as healthcare-system quality indices, governance indicators, or conflict and disaster data, to test whether they add explanatory power beyond the factors already examined.
- Develop an interactive dashboard (e.g. using Power BI or Tableau) allowing policymakers and analysts to explore life-expectancy relationships by country, region and year in real time.
- Investigate the BMI variable using an alternative, more targeted imputation or data-collection strategy so that body-mass-related effects on life expectancy can be assessed without dropping the variable entirely.
- Extend the LASSO modeling exercise to the full 12-feature panel-dataset feature set identified in Chapter 6, with a properly held-out validation sample, to compare its performance and feature importance against the cross-sectional working model presented in this report.

8.4 References

- World Health Organisation (WHO) Global Health Observatory (GHO) data repository — <http://www.who.int/gho/en/>
- Kaggle: Life Expectancy (WHO) dataset — <https://www.kaggle.com/kumarajarshi/life-expectancy-who>
- United Nations Sustainable Development Goals — Goal 3: Good Health and Wellbeing
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Seaborn Documentation
- SciPy Documentation
- Scikit-learn Documentation (Linear Regression, Ridge Regression, Lasso Regression, PCA, StandardScaler)
- Analytics Vidhya: “A Complete Tutorial on Ridge and Lasso Regression in Python” — methodology reference used for alpha-selection approach